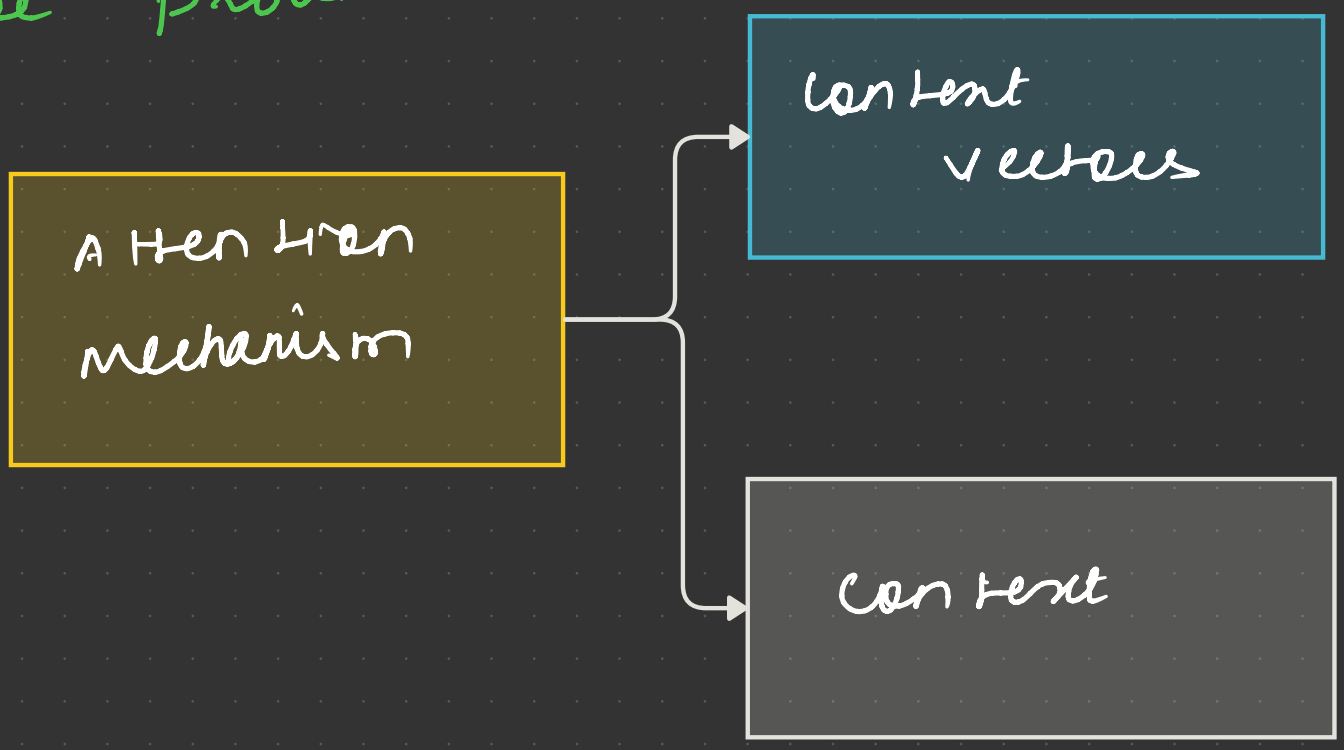
Attention mechanism → Seq2 Seq network

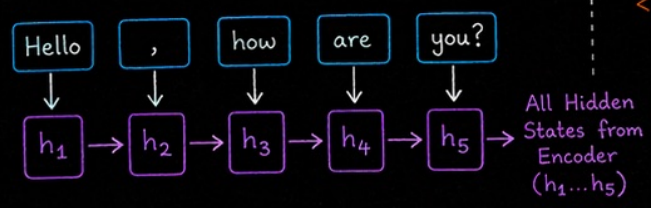
The main idea of the attention mechanism was that for longer or shorter sentence/ Paragraph, along with the context vectors, some more context will be provided.



Attention Mechanism in Seq2Seq

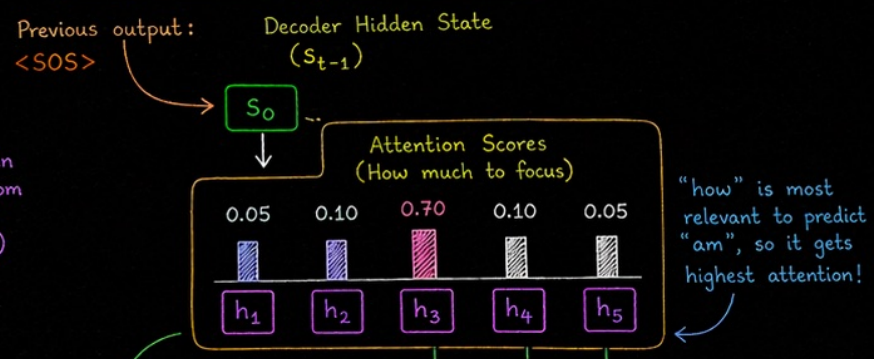
① **Encoder:** Reads Input and gives all hidden states.

Input Sentence:
"Hello, how are you?"

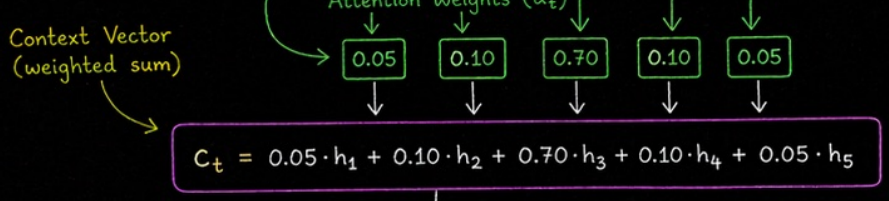


② **Decoder:** At each step, looks at all encoder hidden states and pays ATTENTION to the most relevant words.

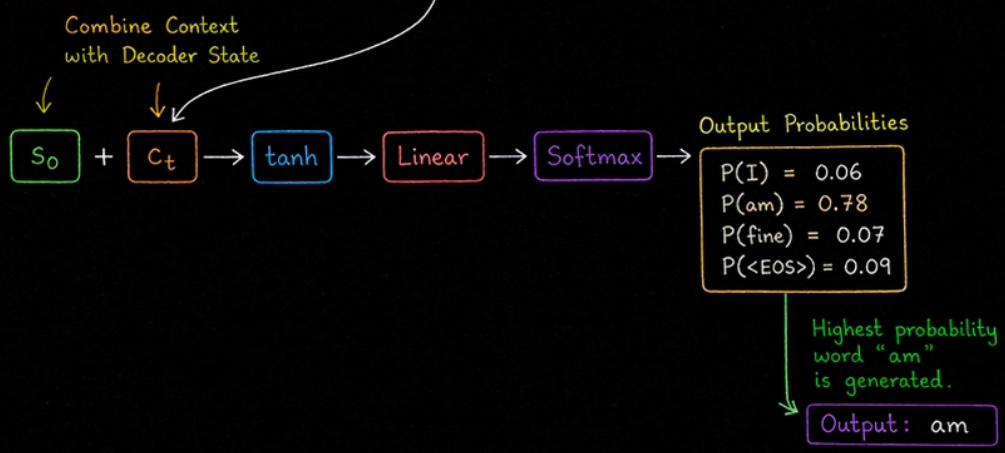
Example: Generating the word "am"



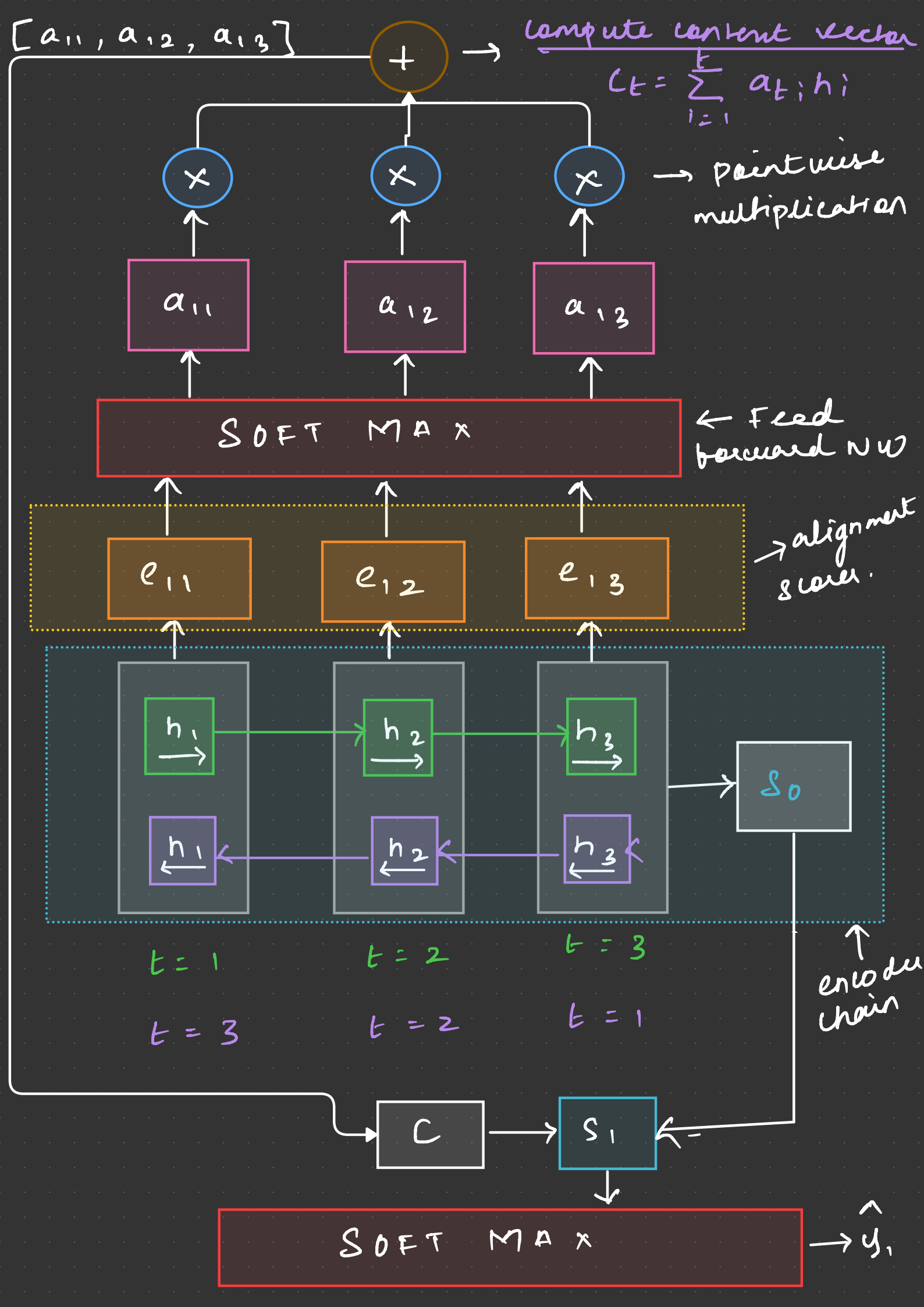
③ **Compute Context Vector:**
Take weighted sum of all encoder hidden states using attention weights.



④ **Generate Output:**
Combine context vector with decoder state and produce next word using Softmax.



Key Idea:
Instead of using only the final encoder hidden state, Attention lets the decoder focus on different parts of the input at each step. This helps generate better and more accurate translations.


[a11, a12, a13]
+
compute context vector
$c_t = \sum_{i=1}^{t} a_{ti} h_i$
×
×
×
pointwise multiplication
a11
a12
a13
SOFT MAX
Feed forward NW
e11
e12
e13
alignment score.
h1
h2
h3
h1
h2
h3
S0
t = 1
t = 2
t = 3
t = 3
t = 2
t = 1
encoder chain
C
S1
SOFT MAX
ŷ1